AP Statistics FRQ Practice Pack

Regenerated edition - topic coverage retained; layout reorganized by difficulty; author/publisher branding removed.

Study use only. Problem contexts follow publicly released AP Statistics FRQ templates. Practice with hints/templates; do not use on graded work unless allowed.

How this pack is organized

1. Four topics kept intact (Exploring Data → Collecting Data → Probability & RVs → Inference).
2. Within each topic, problems are ordered by difficulty:
- * Foundation - identify / interpret / short explanation
- Standard - calculation + justification in context
- * Challenge - full multi-step inference or layered probability
3. Substantive stems unchanged (cleaned for OCR/layout only).
4. Part II contains reference answers for every problem (worked solutions for self-check).

Topic map (CED alignment)

Topic	CED units	Typical FRQ skills
1 Exploring Data	Units 1-2	SOCS, outliers, regression interpret, residuals, r^2
2 Collecting Data	Unit 3	Sampling bias, SRS/stratify, experiments, confounding
3 Probability & RVs	Units 4-5	Conditional probability, Binomial/Geometric/Normal, EV
4 Statistical Inference	Units 6-9	CI/tests for p & mu, chi^2, slope inference

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Part I - Practice Problems

Topic 1 Exploring Data

T1-F1 Foundation * Scatterplot language & residual

Template source label (for study tracking): 2017 FRQ1 (wolves)

Researchers studying a pack of gray wolves collected data on length x (meters, nose to tip of tail) and weight y (kilograms). A scatterplot of weight versus length was described as positive, linear, and strong.

(a) Explain what is meant by each of the following for this situation:
(i) Positive
(ii) Linear
(iii) Strong

The least-squares equation is y-hat = -16.46 + 35.02x.

(b) Interpret the slope of the least-squares regression line in context.

(c) One wolf with length 1.4 m had residual -9.67 kg. What was the weight of the wolf?

T1-F2 Foundation * Two-way table proportions & association

Template source label (for study tracking): 2014 FRQ1 (extracurricular)

An administrator sampled 100 students. Residential status (on/off campus) and extracurricular participation are summarized:

Participation	On campus	Off campus	Total

No activities | 9 | 30 | 39
One activity | 17 | 25 | 42
Two or more | 7 | 12 | 19
Total | 33 | 67 | 100

(a) Calculate the proportion of on-campus students who participate in no extracurricular activity, and the same proportion for off-campus students.

(b) Using the segmented bar graph idea (relative frequencies within each residential status), write a few sentences summarizing the association between residential status and level of participation in the sample.

T1-S1 Standard ** Five-number summary & two outlier rules

Template source label (for study tracking): 2021 FRQ1 (hospital stay)

Length of stay (days) for a random sample of 50 patients:

Days | 5 | 6 | 7 | 8 | 9 | 12 | 21
Patients | 4 | 13 | 14 | 11 | 6 | 1 | 1

(a) Determine the five-number summary of the distribution of length of stay.

(b) Method A = 1.5 x IQR rule; Method B = 2 standard deviations from the mean.
(i) Using Method A, identify potential outliers. Justify.
(ii) Mean = 7.42 days, SD = 2.37 days. Using Method B, identify potential outliers. Justify.

(c) Explain why Method A might identify more outliers than Method B for a distribution that is strongly skewed right.

T1-S2 Standard ** Histogram SOCS, outliers, boxplot vs histogram

Template source label (for study tracking): 2019 FRQ1 (room sizes)

Room sizes (sq ft) for 20 rooms are shown in a histogram (bimodal, roughly 100-350). Summary statistics:

Mean 231.4, SD 68.12, Min 134, Q1 174, Median 253.5, Q3 292, Max 315.

(a) Based on the histogram, write a few sentences describing the distribution of room size.

(b) Determine whether there are potential outliers. Then sketch a boxplot of room size.

(c) What characteristic of the shape is apparent from the histogram but not from the boxplot?

T1-S3 Standard ** Describe distribution; mean vs median under a change

Template source label (for study tracking): 2016 FRQ1 (tips)

A histogram shows Robin’s 60 tip amounts for one day (right-skewed, roughly \$0-\$25).

(a) Write a few sentences describing the distribution of tip amounts.

(b) One tip was \$8. If that tip had been \$18 instead, what effect would the change have on the mean and on the median? Justify.

T1-S4 Standard ** Slope, r^2, largest residual

Template source label (for study tracking): 2022 FRQ1 (bullfrogs)

A biologist fits a least-squares line for predicting bullfrog mass (g) from length (mm):
mass = -546 + 6.08(length), with r^2 ~= 0.819. (Plot of 11 bullfrogs with the line is given.)

(b) Identify and interpret the slope in context.

(c) Interpret r^2 ~= 0.819 in context.

(d) From the residual pattern on the plot:
(i) Approximately what is the length and mass of the bullfrog with the largest absolute residual?
(ii) Does the LSRL overestimate or underestimate that bullfrog’ s mass? Explain.

T1-S5 Standard ** Intercept, r^2, outlier on scatterplot

Template source label (for study tracking): 2018 FRQ1 (checkout)

A random sample of 11 customers: x = customers ahead in line, y = checkout time (seconds).
Computer output: Constant 72.95; Customers in line 174.40; R-Sq = 73.33%; S = 200.01.

(a) Identify and interpret the intercept estimate in context.

(b) Identify and interpret r^2 in context.

(c) One point is an outlier. Circle it on the scatterplot and explain why it is an outlier.

Topic 2 Collecting Data

T2-F1 Foundation * Random assignment to two equal groups

Template source label (for study tracking): 2009 FRQ3 (frog labs)

A teacher will compare physical frog dissection vs computer simulation using 24 students. Analyze posttest - pretest score changes.

(a) Describe a method for assigning the 24 students to two groups of equal size that allows a statistically valid comparison of the two programs.

T2-F2 Foundation * Identify treatments, units, response; control; random assignment

Template source label (for study tracking): 2019 FRQ2 (fungus)

Researchers create four fungus concentrations: 0, 1.25, 2.5, 3.75 ml/L. Equal numbers of insects are placed in 20 containers; each container is sprayed with one mixture. Record number still alive after one week.

(a) Identify treatments, experimental units, and response variable.

(b) Does the experiment have a control group? Explain.

(c) Describe how to randomly assign treatments so each treatment has the same number of units.

T2-S1 Standard ** Matched pairs vs CRD

Template source label (for study tracking): 2022 FRQ2 (acne twins)

A dermatologist recruits 36 pairs of identical twins. Each person receives new drug or placebo for 2 weeks; improvement scored 0-100. Twins have similar acne severity; severity differs across pairs.

(a) Identify treatments, experimental units, and response variable.

(b) Describe a statistical advantage of a matched-pairs design (twins paired) rather than a completely randomized design.

T2-S2 Standard ** Median vs mean; voluntary response vs SRS

Template source label (for study tracking): 2014 FRQ4 (alumni income)

At a reunion, mean income of 50 attendees was reported as the class mean income.

(a) Statistical advantage of using the median of reported incomes rather than the mean as a typical-income estimate?

(b) Method 1: email all 6,826 alumni (expect >=600 responses). Method 2: SRS and phone until all selected respond (about 100 possible). Which method for estimating average yearly income of all class members? Compare.

T2-S3 Standard ** Convenience, SRS, stratification

Template source label (for study tracking): 2013 FRQ2 (campus survey)

Estimate proportion of 70,000 students satisfied with campus appearance; sample size 500.

(a) Why might surveying the first 500 students entering a football stadium be biased?

(b) Describe how to select an SRS of 500 from the list of 70,000 names.

(c) When would stratification by campus give a more precise estimate than stratification by gender?

T2-S4 Standard ** Observational study & confounding

Template source label (for study tracking): 2016 FRQ3 (smoking & Alzheimer’s)

A study tracked 21,123 people for 23 years. For those smoking >=2 packs/day, Alzheimer’s risk was 2.57x that of nonsmokers. Headline: “Strong Association Between Smoking and Alzheimer’s.”

(a) Identify explanatory and response variables.

(b) Observational study or experiment? Explain.

(c) Explain how exercise status could be a confounding variable.

T2-C1 Challenge *** Experiment scope & non-random assignment

Template source label (for study tracking): 2011B FRQ2 (acrophobia)

27 people: 17 randomly assigned D-cycloserine + 2 therapy sessions; 10 get placebo + 2 sessions. Evaluated after 3 months.

(a) Experiment or observational study? Explain.

(b) Drug group improved significantly more than placebo. Can researchers conclude drug + 2 sessions is as beneficial as 8 sessions without the pill? Justify.

(c) If therapists chose who got the drug (no randomization), why might that lead to an incorrect conclusion?

Topic 3 Probability & Random Variables

T3-S1 Standard ** Joint, union, conditional; independence

Template source label (for study tracking): 2019 FRQ3 (medicine adherence)

Relative frequencies:

	Never	Sometimes	Always	Total
Men	0.0564	0.2016	0.2120	0.4700
Women	0.0636	0.1384	0.3280	0.5300
Total	0.1200	0.3400	0.5400	1.0000

One person selected at random.
(a)(i) P(never and woman)
(a)(ii) P(never or woman)
(a)(iii) P(never | woman)
(b) Are “never” and “woman” independent? Justify.

T3-S2 Standard ** Combinations & flawed simulation

Template source label (for study tracking): 2014 FRQ2 (convention)

6 men and 3 women; 3 selected at random; all 3 selected were women.

(a) P(selecting 3 women).
(b) Based on (a), reason to doubt the manager’s “random” claim?
(c) Simulation: roll 3 dice; 1-4 = man, 5-6 = woman; count triple-woman trials. Does this correctly simulate selecting 3 women from 9 people (6M, 3W)? Why/why not?

T3-S3 Standard ** Discrete RV: P, E(X), conditional

Template source label (for study tracking): 2015 FRQ3 (ATMs)

X = number of working ATMs when the mall opens. P(X=0)=0.15, P(1)=0.21, P(2)=0.40, P(3)=0.24.

(a) P(at least one working)
(b) E(X)
(c) P(all three working | at least one working)
(d) Given at least one working, is E(X | X>=1) less than, equal to, or greater than E(X) from (b)? Explain.

T3-S4 Standard ** Binomial definition, P(at least one), expected value

Template source label (for study tracking): 2021 FRQ3 (gift cards)

200 employees; each week one chosen uniformly at random (with replacement across weeks; independent).

(a) For a particular employee in 52 weeks:
(i) Define the RV and its distribution.
(ii) P(at least one gift card).

(b) Expected number of gift cards for a particular employee in 52 weeks; interpret.

(c) An employee gets none all year - strong evidence selection was not random?

T3-S5 Standard ** Geometric wait; binomial count; surprise

Template source label (for study tracking): 2011B FRQ3 (upgrades)

P(upgrade) = 0.10 each flight, independent.

(a) P(first upgrade occurs after the third flight).
(b) P(exactly 2 upgrades in next 20 flights).
(c) In 104 flights, would more than 20 upgrades be surprising? Justify.

T3-C1 Challenge *** Sequential testing; rare event reasoning

Template source label (for study tracking): 2016 FRQ4 (igniters)

Overall igniter failure rate claimed 15%. Test super igniters until first failure or 32 successes.

(a) If failure rate is 15%, P(first 30 succeed).
(b) Given first 30 succeed, P(first failure on 31st or 32nd).
(c) Given first 30 succeed, is it reasonable to believe failure rate < 15%? Explain.

T3-C2 Challenge *** Normal underfill + Binomial crate rejection

Template source label (for study tracking): 2022 FRQ3 (shampoo)

A ~ N(0.60 L, 0.04 L). Underfilled if A < 0.50. Boxes of 10; crate rejected if a random box has >=2 underfilled.

(a) P(a bottle underfilled).

(b)(i) Define RV for warehouse manager and its distribution.
(b)(ii) P(crate rejected).

(c) Adjusted program: N(0.56, 0.03). Recommend original or adjusted? Justify.

T3-C3 Challenge *** Normal percentile, claim probability, expected gain

Template source label (for study tracking): 2019 FRQ5 (battery warranty)

Battery life ~ N(30 months, 8 months). $50 warranty replaces phone if life < 24 months.

(a) In how many months is it expected that 25% of batteries no longer work?

(b) P(a warranty customer requires replacement within 24 months).

(c) Gain +$50 if no claim; loss -$150 if claim. Expected gain per warranty.

T3-C4 Challenge *** Law of total probability; Bayes; Binomial

Template source label (for study tracking): 2018 FRQ3 (left-handed)

3.5% of births are multiple births. Of multiple-birth children, 22% left-handed; of single-birth, 11% left-handed.

(a) P(randomly selected child is left-handed).
(b) P(multiple birth | left-handed).
(c) For a random sample of 20 children, P(at least 3 left-handed).

Topic 4 Statistical Inference

T4-S1   Standard **   One-proportion z-interval + claim about 0.5

Template source label (for study tracking): 2022 FRQ4 (streaming)

Random sample of 920 U.S. teens; 59% use a video streaming service every day.

(a) Construct and interpret a 95% CI for the true proportion.

(b) Based on the CI, do the data provide convincing evidence the true proportion is not 0.5? Justify.

T4-S2   Standard **   One-prop CI → cost interval

Template source label (for study tracking): 2017 FRQ2 (water cups)

Of 80 random customers who asked for a water cup, 23 filled it with soda.

(a) Construct and interpret a 95% CI for the true proportion who do this.

(b) Each soda-fill costs $0.25. If 3,000 ask for water cups in June, use the CI to estimate June’s cost.

T4-S3   Standard **   CI for regression slope

Template source label (for study tracking): 2019 Intl FRQ2 (house prices)

n=20 houses. Regression of price (thousands $) on distance (miles): slope b=-2.158, SE_b=0.149. Conditions met.

(a) Construct and interpret a 95% CI for the true slope (t*~=2.101, df=18).

(b) Agent believes price drops about $2,000 per mile (slope -2). Does the CI contradict that belief?

T4-S4   Standard **   Proportions vs counts; homogeneity hypotheses

Template source label (for study tracking): 2021 FRQ5 (soft drinks)

Independent city samples (Yes = consumed soft drink in past week):

	Baltimore	Detroit	San Diego	Total
Yes	727	1232	1482	3441
No	177	431	798	1406
Total	904	1663	2280	4847

(a) A researcher claims Baltimore has the lowest likelihood of Yes because 727 is the smallest Yes count. Correct? Explain.

(b)(ii) Which city has the smallest Yes proportion? Give the value.

(c)(i) Appropriate inference procedure to compare Yes proportions across cities.
(c)(ii) Hypotheses.

T4-C1   Challenge ***   One-proportion z-test + Type I/II

Template source label (for study tracking): 2021 FRQ4 ($10 coupon)

Manager believes with a $10 coupon, more than 40% of past customers will reorder. Of 90 randomly selected past customers, 38 reorder. alpha=0.05.

(a) Is there convincing evidence the manager is correct? Complete the inference procedure.

(b) Based on your conclusion, which error (Type I or II) could have been made? Interpret in context.

T4-C2 Challenge *** Two-proportion z-test for an increase

Template source label (for study tracking): 2019 FRQ4 (kochia)

2014: 19.7% of 61 plants resistant. 2017: 38.5% of 52 plants resistant. At alpha=0.05, convincing evidence the proportion resistant increased?

T4-C3 Challenge *** Causation from design + two-sample t

Template source label (for study tracking): 2018 FRQ4 (ACL surgery)

210 patients randomly assigned to standard or new ACL procedure.

(a) Would a significant result allow concluding the new procedure causes reduced recovery for patients like these? Explain.

(b) Standard: n=110, x=217, s=34. New: n=100, x=186, s=29.
Convincing evidence new has smaller mean recovery?

T4-C4 Challenge *** Chi-square association

Template source label (for study tracking): 2013 FRQ4 (fruits/vegetables)

Random sample of 8,866 adults: age group vs whether they eat >=5 servings of fruits/vegetables daily.

Age	Yes	No	Total
18-34	231	741	972
35-54	669	2242	2911
55+	1291	3692	4983
Total	2191	6675	8866

Convincing evidence of association between age-group and 5+ servings/day?

T4-C5 Challenge *** Paired t-test

Template source label (for study tracking): 2014 FRQ5 (car prices)

8 car models: one woman and one man bought the same equipped model. Differences (woman - man): mean d-bar = $585, s_d = $530.71.

Do the data provide convincing evidence that, on average, women pay more than men for the same car model?

T4-C6 Challenge *** Chi-square goodness-of-fit

Template source label (for study tracking): 2015 Intl FRQ4 (bank waits)

Acceptable wait-time probabilities: 0.30, 0.25, 0.20, 0.15, 0.10.
Observed counts in n=100: 25, 21, 21, 20, 13.

Conduct a test: are waiting times inconsistent with the acceptable probabilities?

Part II - Reference Answers

Use after attempting. Score yourself on: correct method, work shown, and context in the conclusion.

T1-F1 Scatterplot language & residual

Difficulty: Foundation *

(a) (i) Positive: as length increases, weight tends to increase.
(ii) Linear: the pattern of points is well described by a straight line.
(iii) Strong: points are tightly clustered about the line (little scatter).

(b) For each additional 1 meter of length, the predicted weight increases by about 35.02 kg.

(c) y-hat = -16.46 + 35.02(1.4) = 32.568.
Actual y = y-hat + residual = 32.568 - 9.67 = 22.90 kg.

T1-F2 Two-way table proportions & association

Difficulty: Foundation *

(a) On campus: 9/33 ~= 0.273.
Off campus: 30/67 ~= 0.448.

(b) Off-campus students have a higher proportion with no activities; on-campus students have higher proportions with one activity and with two or more. Residential status and participation level appear associated in this sample.

T1-S1 Five-number summary & two outlier rules

Difficulty: Standard

(a) Min = 5, Q1 = 6, Median = 7, Q3 = 8, Max = 21.

(b)(i) IQR = 2. Fences: 6 - 1.5(2) = 3 and 8 + 1.5(2) = 11.
Potential outliers: 12 and 21 (both > 11).

(b)(ii) Interval: 7.42 ± 2(2.37) = (2.68, 12.16).
Potential outlier: 21 only.

(c) Right skew pulls the mean and SD upward, pushing the 2-SD upper cutoff farther out. Quartiles/IQR are resistant, so the 1.5xIQR upper fence stays lower and can flag more high values.

T1-S2 Histogram SOCS, outliers, boxplot vs histogram

Difficulty: Standard

(a) The distribution is bimodal (peaks near lower and upper ranges), spanning about 100-350 sq ft, with no extreme outliers obvious from the histogram. Center is near the mid-200s; spread is moderately large.

(b) IQR = 118. Fences: 174 - 1.5(118) = -3 and 292 + 1.5(118) = 469.
Min/Max within fences → no outliers.
Boxplot: whiskers 134-315; box Q1=174 to Q3=292; median 253.5.

(c) Bimodality (two peaks) - visible in the histogram, not in the boxplot.

T1-S3 Describe distribution; mean vs median under a change

Difficulty: Standard

(a) Tip amounts are strongly right-skewed, centered near about $5-$10, ranging from about $0 to $25, with a few larger tips in the right tail.

(b) Mean increases (total increases by $10).
Median is unchanged (n = 60; changing one value that stays on the same side of the center does not move the middle order statistics).

T1-S4 Slope, r^2, largest residual

Difficulty: Standard

(b) Slope = 6.08: for each additional 1 mm of length, predicted mass increases by about 6.08 g.

(c) About 81.9% of the variation in bullfrog mass is explained by the linear relationship with length.

(d)(i) Identify the point farthest vertically from the line (largest |residual|) from the plot.
(d)(ii) If the point is above the line, the model underestimates; if below, it overestimates. (On the source plot, the largest |residual| is above the line → underestimate.)

T1-S5 Intercept, r^2, outlier on scatterplot

Difficulty: Standard

(a) Intercept ~= 72.95: when there are 0 customers ahead, predicted checkout time is about 72.95 seconds (may be extrapolation).

(b) r^2 = 73.33%: about 73.33% of the variation in checkout time is explained by the linear relationship with number of customers in line.

(c) The outlier is the point that does not follow the linear pattern (large residual and/or unusual combination of x and y relative to the others).

T2-F1 Random assignment to two equal groups

Difficulty: Foundation *

Label students 1-24. Use a random number generator (or shuffle slips) to select 12 distinct numbers for Group A (physical dissection); the remaining 12 are Group B (simulation). Equal size + chance assignment supports a fair comparison.

T2-F2 Identify treatments, units, response; control; random assignment

Difficulty: Foundation *

(a) Treatments: 0, 1.25, 2.5, 3.75 ml/L.
Units: the 20 containers.
Response: number of insects alive after one week.

(b) Yes - 0 ml/L is a control (no fungus).

(c) Label containers 1-20; randomly permute. Assign first 5 to 0, next 5 to 1.25, next 5 to 2.5, last 5 to 3.75 (5 each).

T2-S1 Matched pairs vs CRD

Difficulty: Standard

(a) Treatments: new drug, placebo.
Units: the 72 people (or 36 twin pairs as matched units).
Response: improvement score 0-100.

(b) Pairing controls for genetic/severity similarity within twins, reducing variability and making treatment differences easier to detect than in a CRD.

T2-S2 Median vs mean; voluntary response vs SRS

Difficulty: Standard

(a) Median is resistant to skew/outliers; income is often right-skewed, so the median better represents a “typical” income.

(b) Prefer Method 2. Method 1 is voluntary response/nonresponse biased (e.g., higher earners more likely to reply). Method 2 is smaller but more likely representative/unbiased.

T2-S3 Convenience, SRS, stratification

Difficulty: Standard

(a) Football attendees are not representative of all students; opinions about grounds may correlate with who attends games → bias.

(b) Number students 1-70,000; use RNG to select 500 unique numbers; survey those students.

(c) When satisfaction differs substantially between campuses (and is more homogeneous within campus) than between genders.

T2-S4 Observational study & confounding

Difficulty: Standard

(a) Explanatory: smoking amount (e.g., packs/day). Response: developing Alzheimer’s.

(b) Observational - no treatment assigned; medical histories were tracked.

(c) Exercise may be associated with smoking and with Alzheimer’s risk, so its effect can be mixed with smoking’s effect.

T2-C1 Experiment scope & non-random assignment

Difficulty: Challenge *

(a) Experiment - treatments randomly assigned.

(b) No - the study never compared to an 8-session no-drug group; conclusions are limited to treatments actually compared.

(c) Selection bias/confounding: therapists might assign the drug to milder cases (or vice versa), mixing patient differences with treatment effects.

T3-S1 Joint, union, conditional; independence

Difficulty: Standard

(a)(i) 0.0636
(a)(ii) 0.1200 + 0.5300 - 0.0636 = 0.5864
(a)(iii) 0.0636 / 0.5300 = 0.12
(b) Yes - P(never|woman) = 0.12 = P(never).

T3-S2 Combinations & flawed simulation

Difficulty: Standard

(a) C(3,3)/C(9,3) = 1/84 ~= 0.0119.

(b) Yes - about 1.2% is unusually small if selection were random.

(c) No - dice model allows repeats and treats trials as independent with P(woman)=1/3 each roll; true selection is without replacement.

T3-S3 Discrete RV: P, E(X), conditional

Difficulty: Standard

(a) 1 - 0.15 = 0.85
(b) 0(0.15)+1(0.21)+2(0.40)+3(0.24) = 1.73
(c) 0.24/0.85 ~= 0.282
(d) Greater - conditioning removes X=0, shifting the mean up (~=2.035).

T3-S4 Binomial definition, P(at least one), expected value

Difficulty: Standard

(a)(i) X = # cards for that employee in 52 weeks; X ~ Bin(52, 1/200).
(a)(ii) P(X>=1) = 1 - (199/200)^52 ~= 0.229.

(b) E(X) = 52/200 = 0.26: over many years, about 0.26 cards per 52-week year on average.

(c) No - P(X=0) ~= 0.771, so getting none is common under randomness.

T3-S5 Geometric wait; binomial count; surprise

Difficulty: Standard

(a) First three not upgraded: 0.9^3 = 0.729.

(b) C(20,2)(0.1)^2(0.9)^18 ~= 0.285.

(c) X ~ Bin(104,0.1), mu=10.4, sigma~=3.06. P(X>20) is tiny (~=0.0005) → yes, surprised.

T3-C1 Sequential testing; rare event reasoning

Difficulty: Challenge *

(a) 0.85^30 ~= 0.00763.

(b) 0.15 + 0.85(0.15) = 0.2775.

(c) Yes - under 15% failure, P(>=30 successes) is very small (~0.0076), so observing 30 successes suggests a lower failure rate.

T3-C2 Normal underfill + Binomial crate rejection

Difficulty: Challenge *

(a) z = (0.50-0.60)/0.04 = -2.5 → P ~= 0.0062.

(b)(i) X = # underfilled in the box of 10; X ~ Bin(10, p~=0.0062).
(b)(ii) P(X>=2) = 1-P(0)-P(1) ~= 0.0017.

(c) Prefer original - adjusted mean raises underfill p (~=0.0228) and rejection probability (~=0.021).

T3-C3 Normal percentile, claim probability, expected gain

Difficulty: Challenge *

(a) 25th percentile: z~=-0.6745 → 30 - 0.6745(8) ~= 24.6 months.

(b) P(X<24) = P(Z < -0.75) ~= 0.2266.

(c) E = 50(1-0.2266) + (-150)(0.2266) ~= $4.67.

T3-C4 Law of total probability; Bayes; Binomial

Difficulty: Challenge *

(a) P(L) = 0.035(0.22) + 0.965(0.11) = 0.11385.

(b) P(M|L) = 0.035(0.22)/0.11385 ~= 0.0676.

(c) X ~ Bin(20, 0.11385); P(X>=3) ~= 0.402.

T4-S1 One-proportion z-interval + claim about 0.5

Difficulty: Standard

(a) phat=0.59. CI: 0.59 ± 1.96sqrt(0.59 0.41/920) ~= (0.558, 0.622).
We are 95% confident the true proportion of U.S. teens who stream daily is between about 0.558 and 0.622.

(b) Yes - 0.5 is not in the interval.

T4-S2 One-prop CI → cost interval

Difficulty: Standard

(a) phat=23/80=0.2875. CI ~= (0.188, 0.387).
95% confident the true proportion is between about 0.188 and 0.387.

(b) Cost = 3000x0.25xproportion → about ($141, $290).

T4-S3 CI for regression slope

Difficulty: Standard

(a) -2.158 ± 2.101(0.149) ~= (-2.471, -1.845).

95% confident the true slope is between about -2.471 and -1.845 (thousands of $ per mile).

(b) No - -2 is inside the interval.

T4-S4 Proportions vs counts; homogeneity hypotheses

Difficulty: Standard

(a) Incorrect - must compare proportions, not counts. Baltimore’s proportion 727/904 ~= 0.804 is actually the largest.

(b)(ii) San Diego: 1482/2280 = 0.65.

(c)(i) Chi-square test of homogeneity.
(c)(ii) H0: Yes probability is the same in all three city teen populations.
Ha: at least one city’s probability differs.

T4-C1 One-proportion z-test + Type I/II

Difficulty: Challenge *

(a) H0: p=0.40 vs Ha: p>0.40.
phat=38/90~=0.422.
z=(0.422-0.40)/sqrt(0.4 0.6/90)~=0.43; p-value~=0.33 > 0.05.
Fail to reject H0 - not convincing evidence that more than 40% will order.

(b) Type II possible: failing to reject a false H0 - concluding the belief isn’t supported when the true rate really is >40%, possibly missing a useful promotion.

T4-C2 Two-proportion z-test for an increase

Difficulty: Challenge *

H0: p_2017 = p_2014 vs Ha: p_2017 > p_2014.
Using counts ~=12/61 and 20/52: pooled phat_c~=0.283, z~=2.21, p-value~=0.014 < 0.05.
Yes - convincing evidence of an increase.

T4-C3 Causation from design + two-sample t

Difficulty: Challenge *

(a) Yes - random assignment supports a causal conclusion for patients like those studied.

(b) Ha: mu_new < mu_std.
t=(186-217)/sqrt(34^2/110+29^2/100)~=-7.13; p~=0.
Yes - convincing evidence of shorter mean recovery for the new procedure.

T4-C4 Chi-square association

Difficulty: Challenge *

Chi-square test of independence/association.
chi^2 ~= 8.98, df=2, p~=0.011 < 0.05.
Yes - convincing evidence of association.

T4-C5 Paired t-test

Difficulty: Challenge *

Paired t-test: H0: mu_d = 0 vs Ha: mu_d > 0.
t = 585 / (530.71 / sqrt(8)) ~= 3.12; df = 7; p ~= 0.008 < 0.05.
Yes - convincing evidence women pay more on average.

T4-C6 Chi-square goodness-of-fit

Difficulty: Challenge *

chi^2 GOF: H0: proportions equal claimed; Ha: at least one differs.
Expected: 30, 25, 20, 15, 10.
chi^2 ~= 4.09, df=4, p~=0.39 > 0.05.
Fail to reject - data are not inconsistent with the acceptable probabilities.

Suggested practice path

1. Finish all * in Topics 1-2 (speed + template language).
2. Finish across all topics (exam pacing).
3. Save * inference/probability for timed sets (25-45 minutes).
4. Rewrite conclusions from memory without looking at Part II.

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